Memo: Completeness of relational operations in LLMs

From: Daryl Alfaro **Date:** October 13, 2026 **Re:** preliminary exploration (deliverables from the research brief)

Summary. The phenomenon is real and measurable. On 365 questions whose records were all in the prompt, five models had mean recall of 0.966–0.996, yet only 81–98% of their answers were exactly the correct set. The errors are omissions, not hallucinations, for four of the five models. Completeness falls as answer sets and tables grow, and prompting does not fix it. The two DeepSeek models are near ceiling at the current difficulty, so a full benchmark needs harder tiers. The attached results paper has the details, and the code, data and every raw response are at https://github.com/daryl-888/llm-relational-completeness.

Deliverables against the brief

Brief asked for	Status
Related work	Done; summary below, about 20 papers in `docs/related_work.md`.
Proof of concept	Done. Reproducible failures with known ground truth. For example, DeepSeek V4 Pro was incomplete on the same 50-item list in all four runs, and Claude Haiku 5.5 left Maine out of "U.S. states containing 'in' " in all four runs (paper, Box 1 and the failure-case table).
Preliminary generalization	Done, and broader than asked. Selection, JOIN and aggregation, plus projection, nested queries and set operations. Every operation except set operations produces incomplete answers for at least three models. Wrong totals on COUNT/SUM questions are mostly undercounts (9 of 11).
Benchmark proposal	Built and run rather than only proposed: 500 questions, a task taxonomy of 7 operations × 3 settings, SQL-derived ground truth, and the metrics exact-set accuracy, recall/precision and a missing/near-miss/wrong/made-up/duplicate error taxonomy.
Feasibility assessment	Below.

Closest prior work. *HoloBench* (Maekawa et al., ICLR 2025) is closest. It writes out database rows as long text and evaluates five query types while varying context length and information density, but it scores with a model-based judge and partial credit, and does not control the number of correct items. *Oolong* (Bertsch et al., 2025) covers long-context aggregation with scalar answers. *LongProc* (2025) scores HTML-to-TSV extraction by row-level F1, with no selection conditions. *LOFT* (Lee et al., 2024) finds SQL-like compositional reasoning hard for long-context models. Multi-answer QA (QAMPARI, RoMQA, QUEST) mixes up retrieval, knowledge and reading failures. *Galois* (Saeed et al., EDBT 2024) queries a model's memory with SQL. What is new here:

1. answer-set size is a controlled variable;
2. ground truth comes from executing SQL, with a set-level error taxonomy and no LLM judge;
3. one framework covers seven operations;
4. it reports the reasoning cost of completeness.

Novelty risk: medium. HoloBench is the baseline to compare against directly.

Feasibility verdict: proceed, with changes.

- *Robust?* Yes, as a rate. Two independent samples agree on 87–100% of verdicts, and model differences are statistically clear: three models trail the best by 7–17 points, with intervals excluding zero. Individual failures sometimes recur and sometimes don't, so results should be reported over many questions and several samples.
- *Generalizable?* Across operations, mostly yes. Across access paths, partly: the memory-only slice shows the same omissions, but the retrieval slice as built measures only the retriever.
- *Distinct from existing work?* Yes on answer-size control and set-level error types; that distinction should be shown empirically against HoloBench.
- *Headroom?* Limited for the strongest models: the DeepSeek models are exact on 98% of in-context questions.

MiMo-V2.6 Flash, the weakest, is exact on only 34% of questions with 50 or more correct items.

Recommended changes for a full benchmark

1. **Harder tiers:** answer sets of 200–1,000 items and tables of 5k–20k rows, to restore headroom.
2. **A fully crossed design** of answer size × table size × format. These are partly confounded today (large answers only appear in large tables) and are currently separated by regression.
3. **Control reasoning:** a reasoning-off condition and fixed reasoning budgets, to test whether the reasoning–completeness link is causal.
4. **Redesign the retrieval slice** with queries where the retriever's top results include near-miss distractors.
5. **More rigour:** at least 3 samples per question, more questions per cell (the main trend rests on 100 questions per model), a real-data slice, and frontier models.

Questions for you.

1. Should the next step target a paper (with a direct HoloBench comparison), or first a larger benchmark?
2. Which models should be in scope, and is there a compute or API budget for frontier models?